# Parth Khandagale

PhD. in Chemical Engineering

pkhandag@buffalo.edu +1-716-398-6578 parth khandagale Google Scholar

## ABOUT ME

**5+ years** of research Experience and **1.5+ years** of industrial experience.

Currently engaged in **advanced material research** and **MXene-based nanocomposites** to design lightweight, high-performance **electromagnetic interference (EMI)** and **neutron radiation shielding materials.** Experienced in advanced materials characterization, including **XRD, XPS, Raman, FTIR, XRF, SEM-EDS, TEM, AFM, TGA, DSC, BET** surface-area analysis, **PPMS**, and **UV-Vis spectroscopy**.

**Highlighted Work:**

- **3+ years on MXenes Synthesis**
- **Designed gas-inducing Impellers**
- **Radar Absorbing Materials**
- **RF Shielding**
- **Up-Conversion Photonics**
- **Lanthanide Chemistry**

## EDUCATION

**University at Buffalo (SUNY)** Aug 2023 – Present

**Ph.D**. in Chemical Engineering

GPA: **3.79/4.00** Buffalo, NY

**US - AFOSR** sponsored Projects

**Institute of Chemical Technology** Aug 2018 – Jul 2023

Integrated **M.Tech.** in Chemical Engineering

Minor in Energy Engineering

**66.67%** dye degradation using **Photocatalysis**

GPA: **9.32/10.00** Mumbai, IN

## RECENT PROJECTS

**Electromagnetic Interference and Neutron Radiation Shielding** **January 2025 – December 2025**

- **40 decibels** of Electromagnetic Interference Shielding from **MXene - Exfoliated Boron** binder-free hybrid films.
- **3µm** thin films shield **3%** of thermal neutron radiation.

**Lithographic patterning for dual mode anti-counterfeiting** **January 2025 – December 2025**

- **Optically up-converting** and Electrically conductive **RFID security tags**.
- **Lithographic patterning** to create microscale features on nearly transparent optical and electronic films.

**Exploring etching strategies to synthesize Neodymium MXenes.** **January 2024 – December 2024**

- Lanthanide containing in plane *MAX* phases and MXenes
- Exploring etching strategies to synthesize $\mathbf{(Mo_{2/3}Nd_{1/3})_2CT_x}$

## NOTABLE EXPERIENCE

**Vesuvius – Foseco India Ltd.** **July 2022 – October 2022**

*Project Intern* *Pune, Maharashtra*

∗ **Foundry and foundry consumables**. Developed process **cost modules** to optimize raw materials purchase operations.

∗ Centralized global supplier database to streamline raw commodities imported by the company.

**Snowtech Equipment Pvt. Ltd.** **September 2021 – December 2021**

*Project Intern* *Mumbai, Maharashtra*

∗ **Design and fabrication** of pitched blade and hollow pipe **Gas Inducing Impellers** for gas-liquid contactors.

∗ Estimated **critical impeller speed** and **mixing time** for gas-liquid systems.

**Godrej Industries Ltd.** **September 2019 – December 2019**

*Student Intern* *Valia, Gujarat*

∗ Analysed production of **glycerine** and **fatty acids** using Fat splitting reaction.

∗ **Hydrogen production** using **steam reformation** and Sodium Lauryl Sulphate **surfactant production**.

## Selected Publications

[1] "Rare-Earth-Modified Luminescent Two-Dimensional Nanosheets Derived from Oxidized $Mo_2CT_x$ MXene," ***2D Materials*, 2026**.

[2] "Linear and Nonlinear Optical Response of Rare-Earth i-MAX Carbides," ***ACS Applied Optical Materials*, 2026.**

[3] "Advanced Oxidation-Based Pretreatment of Wastewater for Enhanced Biofuel Production," ***Biofuels*, CRC Press, 2023.**